

DATA SCIENCE FOR CLINICIANS

Garrett Bullock, PT, DPT, DPhil

'Lil' about Me

Currently

- Associate Professor, Wake Forest University School of Medicine
- Director of Statistics for the Musculoskeletal Institute
- Consultant across sport, exercise, and orthopaedic industries
- Practicing sport physical therapist

Background

- Professional baseball player
- Undergrad degree in History (Middle East Concentration)
- Clinic manager in rural outpatient PT clinic
- Came to coding during PT school
 - Motivation to solve my own clinical data driven problems!

Why This Course?

Evidence Based (or Informed) Practice is Becoming Data Centered.



'New World' of Clinical Practice

- Patient reported outcomes
- Patient data extraction (electronic medical record)
- Data dashboards for multiple patients
- Remote Patient monitoring
- Quality improvement
- Learning health systems
- Part of clinician review

'New World' of Clinical Practice

- Patient reported outcomes
- Patient data extraction (electronic medical record)
- Data dashboards for multiple patients
- Remote Patient monitoring
- Quality improvement
- Learning health systems
- Part of clinician review

Clinicians Need Data Skills & Data Language



Potential problems clinicians might want to solve

	Analysis focus	General question framing	Illustrative example
Clinical profiling	Illness trajectory	For individuals diagnosed with <a selected condition>, what characteristics can be identified from their prior clinical records and history?	For people who suffer from a rotator cuff injury, what patterns are observed in age distribution, sex, comorbidities, medication history, and healthcare use?
	Therapy patterns	Among patients with <a selected condition>, which therapeutic options from <a defined treatment set> are used, and how do patients transition between them over time?	In patients who suffer a rotator cuff injury, what medical services concerning physician, physiotherapist, imaging, and prescription medication are used and sequenced across treatment episodes?
	Event frequency	For individuals initiating <a specified medication>, what proportion experience <a known safety or clinical event> within a defined follow-up period?	Among patients who suffer a rotator cuff injury, what proportion undergo rotator cuff reconstruction during the first year of therapy?

Potential problems clinicians might want to solve

Analytics category	Analysis focus	General question framing	Illustrative example
Population-level outcome assessment	Drug safety monitoring	Is exposure to <a specified medication/intervention> associated with a higher likelihood of <a particular adverse outcome> within <a defined timeframe> after treatment initiation?	Does starting physiotherapy for a within 2 weeks following a rotator cuff injury reduce the risk of rotator cuff reconstruction during the year?
	Comparative effectiveness	When compared to <an alternative therapy>, does <a specified medication/intervention> change the risk of <a clinical benefit or harm> during <a defined exposure window>?	Relative to steroid injections, does physiotherapy alter the risk of rotator cuff reconstruction during the first year?

Potential problems clinicians might want to solve

Analytics category	Analysis focus	General question framing	Illustrative example
Individual-level prediction	Disease progression	For a patient newly diagnosed with <a specified disease>, what is the estimated probability of developing <a related condition or complication> within <a given follow-up horizon>?	For a person with newly identified rotator cuff injury, what is the likelihood of undergoing total shoulder arthroplasty within the next three years?
	Therapeutic benefit	For a patient beginning <an intervention>, how likely is it that the patient will achieve <a predefined clinical goal> within <a specified time period>?	For a person with glenohumeral osteoarthritis that suffers a rotator cuff injury, what is the probability of sustaining activities of daily living over the next three years?
	Treatment-related risk	For an individual newly exposed to <a specified therapy>, what is the chance of experiencing <a safety outcome> within <a defined	For a patient undergoing total shoulder arthroplasty, what is the estimated risk of revision within one year?

Why Are We Learning in R?

- Open Source (Free!)
- No Barriers for Use
- Large Community of Users
- Specifically Designed for:
 - Statistics
 - Epidemiology
 - Data Visualization



Pre-Requisites

- Pre-Course Videos
- Downloading R Tools
- Downloading R and R Studio
- Installing and Uploading Packages
- Uploading Data
- 'Tidyverse' & the 'Pipe'



Objectives

1. Adhere strictly to ethical guidelines regarding participant consent, conflicts disclosure, data transparency and framing of patient projections communicated to key knowledge users.
2. Articulate methodology assumptions, strengths, and gaps transparently to allow independent expert scrutiny prior to supplying any data reports or analyses to clinicians, healthcare leaders, and/or researchers.
3. Visualize different data sources into best scientific practice for effective communication to clinicians, healthcare leaders, and/or researchers.

Objectives

4. Identify, communicate, and control for confounding in clinical based questions.

5. Calculate prevalence, incidence, and rates and communicate the different clinical inferences of these epidemiological calculations.

6. Manage uncertainty bias, overfitting limitations, data leakage, and algorithmic bias to improve reliability and generalizability of patient data analyses.

7. Appraise best practices for prediction framework conceptualization spanning research inquiry, methodology, implementation, and responsible communication of patient

Why Are You Here?

- 'Dry Run' of this Course
- So you get it for free!

Throughout This Course

- Provide feedback
- Let me know if I need to change anything
- Everything is open source (code & data)

SOME RECOMMENDATIONS

- Practice writing code 'Analog'
- Have a dedicated hard copy notebook
- This will greatly increase your learning speed
- Will be a great reference for later



SOME RECOMMENDATIONS

- Encourage to 'get your hands dirty'
- Best learning is using real data
- Use Kaggle and other open source data to replicate previous coding blogs
- Gradually move to creating your own practice projects



SOME RECOMMENDATIONS

- Large language models can help streamline learning
- BUT from experience – **DO NOT** generate original code from these sources
- Think of them like a calculator
- Can tweak/refine but not create
- Will become a dark rabbit hole



